

**Analysis Report on
Student Performance**

Vraj Patel

11-18-25

Introduction and Dataset Description

Academic performance is shaped by a combination of factors. Things like habits, routines, and prior experiences are some of these factors, yet the extent of how much each of these contribute to the overall success is not always easy to recognize without careful statistical analysis. Educational research usually points to things like study time, sleep, prior knowledge of something, and enrichment activities as being important influences, but datasets that capture these variables together would allow for a better and more precise understanding of how they interact with each other. In this project, I worked with a dataset of 10,000 students that records multiple behaviors and academic indicators for each individual student. HoursStudied looks at the total number of hours each student dedicated to studying. PreviousScore looks at the results from previous examinations and provides a numerical indication of the student's academic foundation before the current assignment, ExtracurricularActivities indicates if a student was involved in organized activities outside the classroom. SleepHours looks at the average amount of sleep per day, and SampleQuestionPapersPracticed keeps track of how many practice papers the student completed. A preview of the dataset that shows the first several observations and the structure of all the variables, appears in (Appendix 1).. These variables together provide a well-rounded view of both academic input and daily habits, making the dataset suitable for building a regression model that explains and predicts student performance.

The dependent variable for this project was PerformanceIndex, a numerical measure that represented the student's overall performance. It ranges from 10 to 100 and is like a summary score of the academic achievement that a student has had. Understanding what drives this index requires more than just running a regression it requires being able to understand how each predictor behaves and how the predictors interact with each other. Before I started any modeling, I went through a full exploratory phase to get a better grasp of how the dataset behaved in practice. This helped me guide each step of the way that came after and made the actual modeling decisions easier.

Exploratory Analysis

The first part of my exploration focused on the distribution of the PerformanceIndex. I used a histogram to see how the scores were spread across the 10,000 students and whether the overall shape looked normal enough to support a linear model. The histogram showed a smooth curve with most students being clustered around the mid-50s, forming a pattern that was pretty

close to normal (Appendix 2). There were only a few very high OR low performers on the edges. This was a good sign to see because a stable and centered dependent variable usually leads to a more cleaner regression model.

After understanding the outcome variable, I examined how each numerical predictor related to performance using scatterplots. This part of the analysis made the relationships very clear visually. The first scatterplot showed PerformanceIndex compared with HoursStudied (Appendix 3). The points had a bit of an upward trend, but the relationship wasn't super closed yet. Students who studied the same amount sometimes performed way differently, this suggests that while study hours do help, they aren't the most important factor.

The second scatterplot, which compared PerformanceIndex with PreviousScores, resembled a completely different story (Appendix 4). Here, the trend was strong and borderline perfectly linear. Students who had high scores before, tended to do good again and the relationship was so consistent that it stood out right away as a major predictor. This makes sense because since having prior academic achievement usually means there is a better long-term understanding.

The next two scatterplots showed the relationships for SleepHours and SampleQuestionPapersPracticed (Appendix 5 and Appendix 6). Both of these predictors also had positive trends, but way weaker than the one seen with the PreviousScores variable. Students who had slept more or practiced more sample papers, tended to perform slightly better but their points were more spread out. This suggests that these behaviors do help, but again are not as important in predicting performance as study habits or prior scores.

Since extracurricular participation is a categorical variable like 0 or 1, I used a boxplot next to strictly compare the two groups. The boxplot showed that students who participated in extracurricular activities tended to have a bit of a higher median performance and less super low scores (Appendix 7). The difference wasn't that much but it was still noticeable. This suggested that extracurriculars might have a small positive influence. This may have been because students who stay active tend to balance their time better or they stay more engaged overall.

Together, these exploratory visuals gave me a clear picture of the dataset. PreviousScores was clearly the strongest predictor, while HoursStudied, SleepHours, and SampleQuestionPapersPracticed showed smaller but still positive effects. Extracurricular

involvement had a subtle impact as well. Most importantly, all the relationships looked reasonably linear, which supported moving forward with regression modeling.

Regression Analysis and Model Building

After I was done with the exploratory part of the project, What I did next was build the regression model to see how all the predictors would behave when they were considered together. I first started with the full regression model that included each predictor: HoursStudied, PreviousScores, SleepHours, SampleQuestionPapersPracticed, and Extra. The goal here was to see which variables were valuable/important when the others were also included. In the full model output (Appendix 8), every predictor came out significant with very small p-values, meaning all five variables played a meaningful role in predicting the PerformanceIndex.

The model itself had an R-square of 0.9888, this meant that the predictors explained about 99 percent of the variation in student performance! That is very high, and it lined up with what I saw earlier in the scatterplots, especially the strong linear trend with PreviousScores. The parameter estimates also made sense based on the earlier visuals. PreviousScores had the most impact, HoursStudied had a noticeable positive effect, and SleepHours and SampleQuestionPapersPracticed both had smaller but still positive coefficients. Extra also had a positive coefficient which showed that students who participated in extracurricular activities DID tend to perform a little bit better. The final equation from this model was:

$$\text{PerformanceIndex} = -34.076 + 2.853(\text{HoursStudied}) + 1.018(\text{PreviousScores}) + 0.481(\text{SleepHours}) + 0.194(\text{SampleQuestionPapersPracticed}) + 0.613(\text{Extra})$$

This full model showed that each predictor was contributing something to the final performance score. There was nothing that seemed to be out of place, and the variance inflation factors (VIFs) were all very low, which means that multicollinearity was not an issue at all here.

Since the full model showed that everything was significant, I wanted to make sure of this by using a stepwise regression. The stepwise method tests each variable one by one to see if anything should actually be removed based on its contribution to the model. The stepwise output (Appendix 9) ended up selecting the exact same five predictors. In other words, it didn't drop anything. The final step of the stepwise process showed the same R-square of 0.9888, the

same significance pattern, and the same core structure. This confirmed what the full model suggested: all five predictors mattered, and none of them were weak enough to remove.

Since the stepwise selection didn't change anything, the final refined regression model was the exact same as the full model. The coefficients stayed the same, the R-square stayed the same, and the significance levels also were the same. This was actually nice to see because it meant the model was already very stable and well made from the beginning and I realized this the further I went into the project. Sometimes stepwise analysis can reveal redundancy or unnecessary predictors, but in this case, each variable genuinely added something to understanding student performance. The fact that both approaches matched perfectly made it very straightforward to interpret the final model with confidence.

Diagnostics and Prediction Analysis

After confirming the final regression model, the next step was to make sure the model behaved the way it was supposed to. A high R-square is good, but it doesn't mean much if the model's assumptions aren't met. To check this, I used the diagnostic plot generated in my SAS output, which appears in (Appendix 10). The plot compares the studentized residuals of the model to what we would expect if the residuals followed a normal distribution. In the screenshot, the points line up almost perfectly along the diagonal line. Seeing the straight alignment is what you want to see because it means the model's errors are evenly spread out too and they follow a normal pattern. There weren't any big deviations or clusters so there was no sign of skewness or pockets of outliers that could distort the regression. Having a dataset of 10,000 students, if there were small issues, they would have shown up so seeing such a clean line gave me confidence that the model was statistically reliable.

Once I made sure that the model's assumptions were met, I then used the model for actual made up predictions. Using the SAS code I wrote, I tested the model on a new set of input values which were the following: a student with 6 hours studied, a previous score of 85, 7 hours of sleep, 5 sample papers practiced, and participation in extracurricular activities. These inputs correspond to the datalines "6 85 7 5 1" used in the code and match the prediction table shown in (Appendix 11). The model predicted a PerformanceIndex value of roughly 74.55 for that student. This matched my expectations because the earlier scatterplots showed that PreviousScores and HoursStudied were the strongest drivers of performance, and the

diagnostic-confirmed regression equation incorporated those strengths directly into the predicted value.

When you look at the full prediction table, the predicted values and actual dependent values are always close together and the residuals are small. This consistency shows that the model isn't just fitting the training data good, but it is also producing stable predictions for NEW data. The confidence intervals are narrow too which means the model isn't producing estimates that are uncertain. Together, the diagnostics and predictions confirmed that the final refined regression model was functioning exactly as intended and that the relationships among the predictors produced reliable and meaningful predictions for student performance.

Conclusion and Future Work

Bringing everything together, the project showed clearly that all five predictors played a meaningful role in explaining student performance. The strongest relationships came from PreviousScores and HoursStudied, which lined up with both the scatterplots and the regression coefficients. These two variables consistently pushed predicted performance upward in a noticeable way, and their stability across every model made them stand out as the most reliable indicators. The full model and the final refined model both told the same story: these predictors together explained about 99 percent of the variation in the PerformanceIndex. This level of fit is extremely strong, and the diagnostics supported it by showing clean residual patterns with no major problems. The predictions I created also behaved exactly how the model suggested they would by creating values that were close to the actual ones and staying within close confidence intervals. I think that these results confirmed that the model was solid and very capable of producing important insights into what factors really drive academic performance and success for students.

For my future work, one thing I did that could be strengthened is the way I went about handling the categorical variable for extracurricular activities. I converted the Yes/No values into a dummy variable which was the Extra variable and used it directly in the model. I know that this is a normal and the right thing to do, I could have spent more time checking whether different types of activities performed differently. What I mean by this is, if I treated extracurricular involvement as if all activities had different potential impacts. The reality is that being in a sport, a club, or a volunteer organization, they are all unique and they may affect performance differently. I think a better approach would be to break the activity variable into multiple

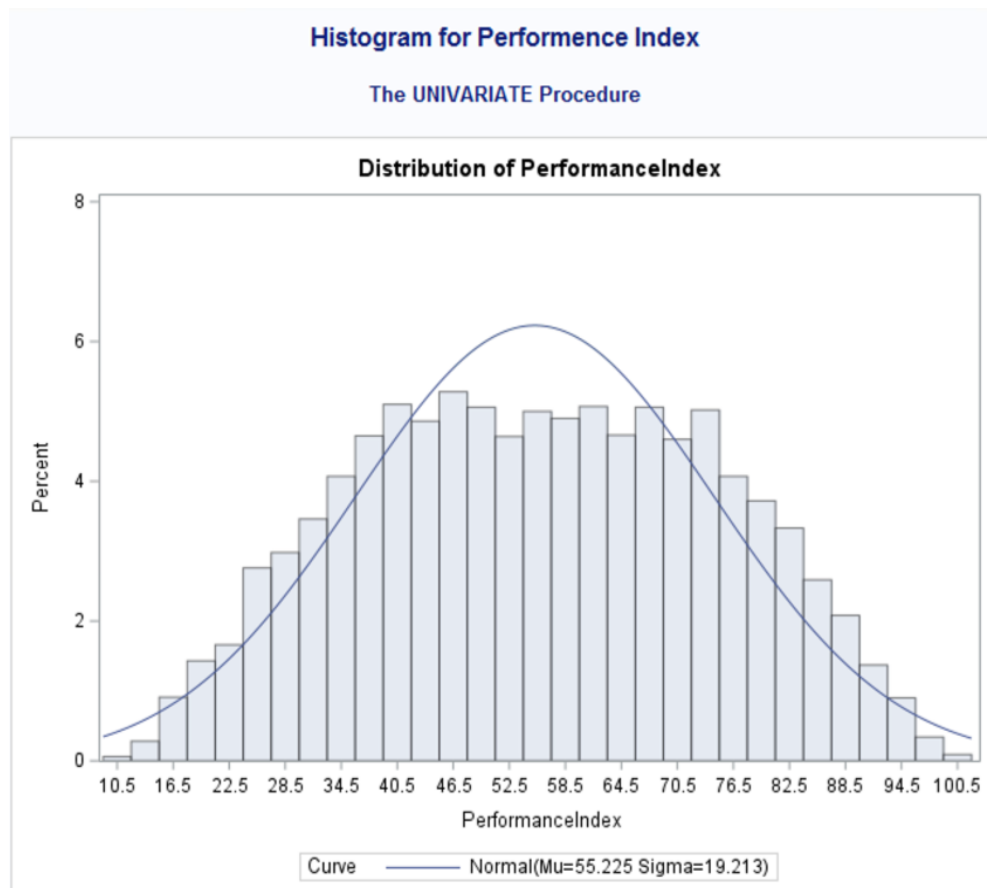
categories if that information was available, obviously in this dataset it was not. This could have allowed me to see whether certain kinds of involvement helped more than others and that would have given a more detailed view of how extracurricular participation contributes/fits into academic success. The reason this would help is because the model would reflect more realistic differences between students instead of assuming all activities influence performance the same way.

Lastly, while I focused on linear relationships, the dataset might contain patterns that are not perfectly linear. Variables like sleep or study hours might have diminishing returns, meaning the benefit could level off at some point. Exploring nonlinear models or interaction terms between predictors could uncover more nuance that the linear model does not capture. Testing interactions, for example, could show whether studying helps more when previous scores are high, or if sample paper practice benefits certain students differently. This would help deepen the understanding of how these habits and academic indicators work together in real-life situations.

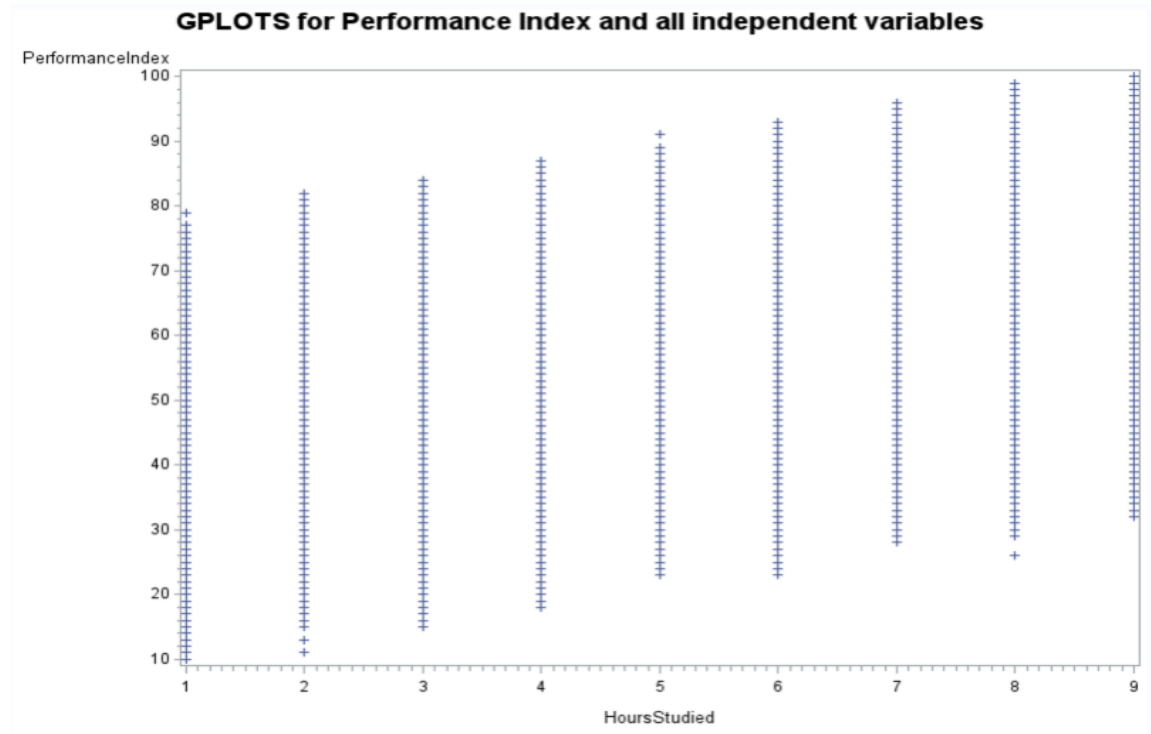
Appendix 1: Preview of Student Performance Dataset

Preview of Student Performance Data							
Obs	HoursStudied	PreviousScores	ExtracurricularActivities	SleepHours	SampleQuestionPapersPracticed	PerformanceIndex	Extra
1	7	99	Yes	9	1	91	1
2	4	82	No	4	2	65	0
3	8	51	Yes	7	2	45	1
4	5	52	Yes	5	2	36	1
5	7	75	No	8	5	66	0
6	3	78	No	9	6	61	0
7	7	73	Yes	5	6	63	1
8	8	45	Yes	4	6	42	1
9	5	77	No	8	2	61	0
10	4	89	No	4	0	69	0
11	8	91	No	4	5	84	0
12	8	79	No	6	2	73	0
13	3	47	No	9	2	27	0
14	6	47	No	4	2	33	0
15	5	79	No	7	8	68	0

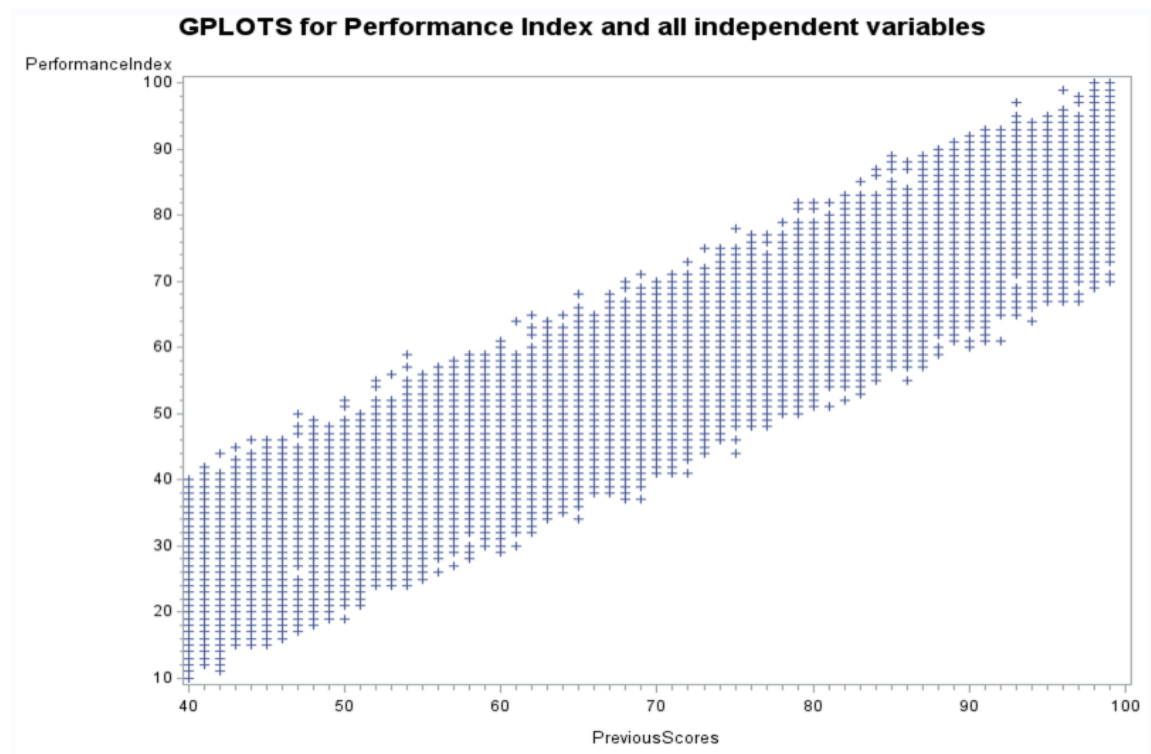
Appendix 2: Histogram for PerformanceIndex



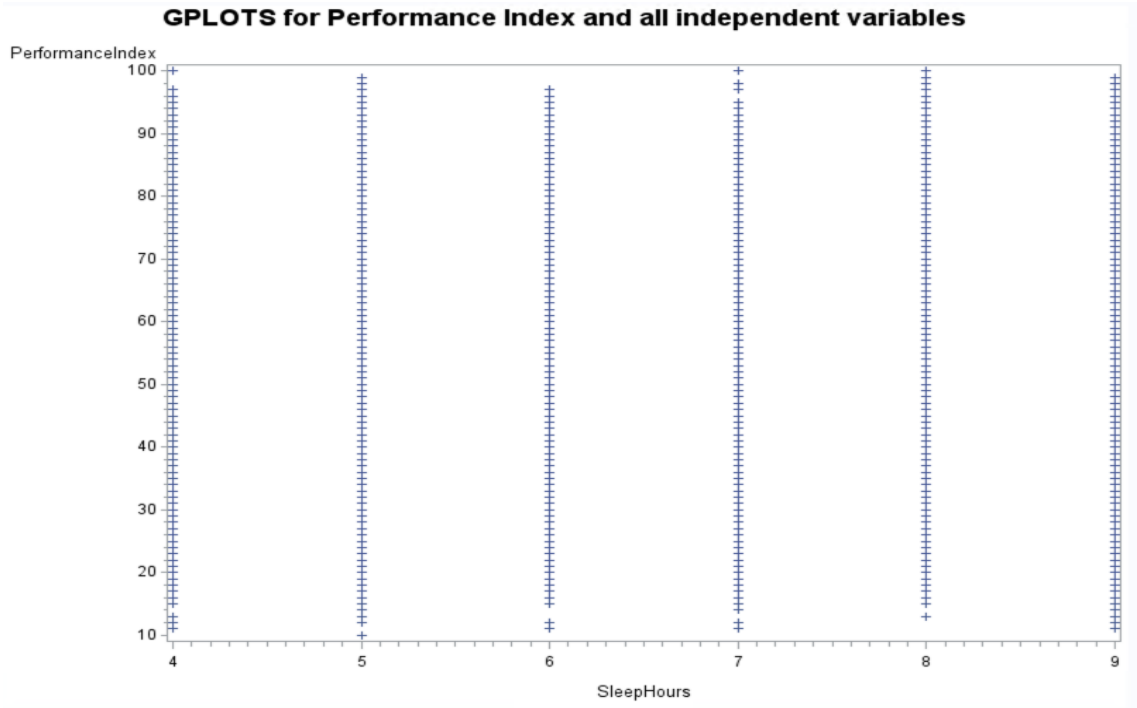
Appendix 3: Scatterplot of PerformanceIndex vs HoursStudied



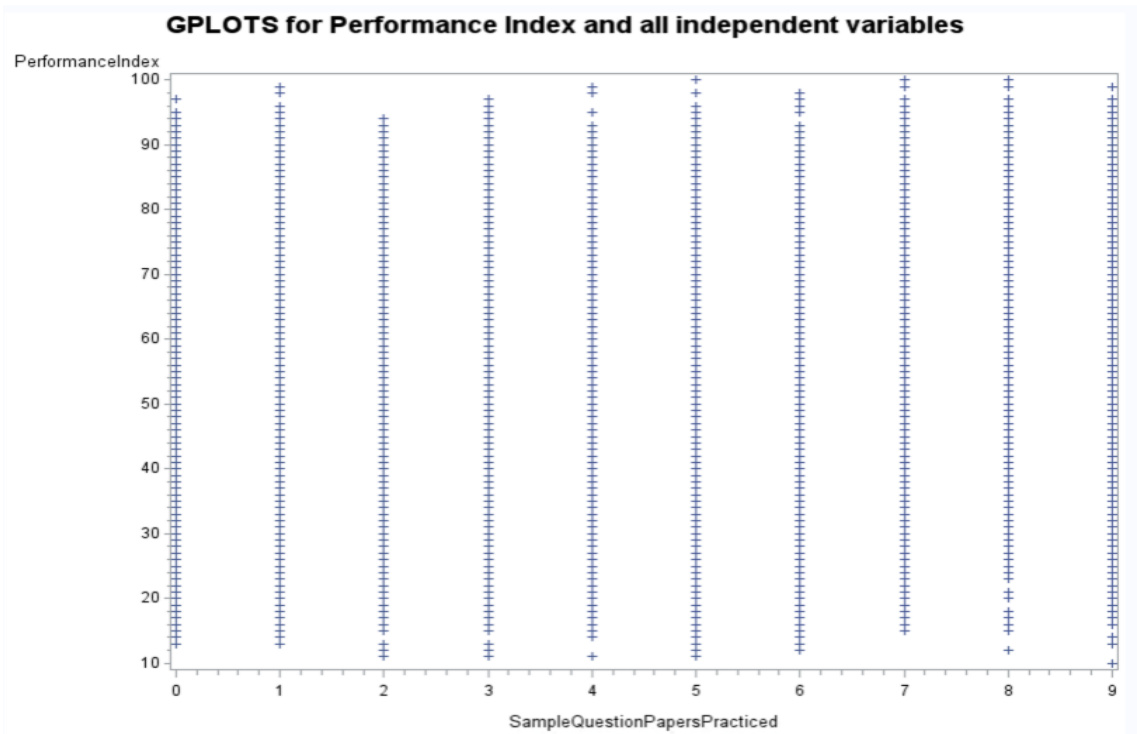
Appendix 4: Scatterplot of PerformanceIndex vs Previous Scores



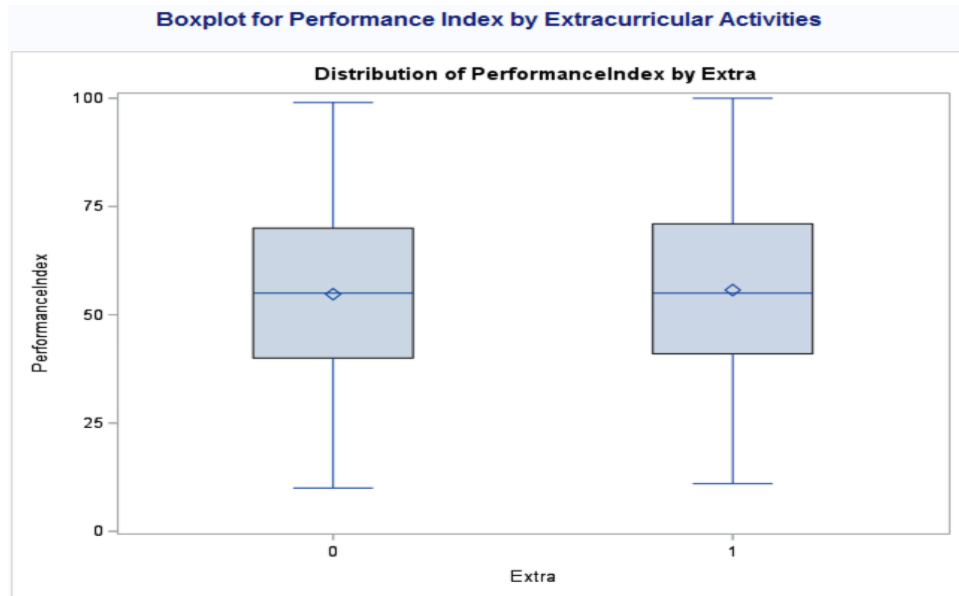
Appendix 5: Scatterplot of PerformanceIndex vs SleepHours



Appendix 6: Scatterplot of PerformanceIndex vs SampleQuestionPapersPracticed



Appendix 7: Boxplot of PerformanceIndex by Extracurricular Activities



Appendix 8: Full Regression Model Output

Full Regression Model for Performance Index with VIF

The REG Procedure

Model: MODEL1

Dependent Variable: PerformanceIndex

Number of Observations Read	10000
Number of Observations Used	10000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	3649341	729868	175709	<.0001
Error	9994	41514	4.15384		
Corrected Total	9999	3690855			

Root MSE	2.03810	R-Square	0.9888
Dependent Mean	55.22480	Adj R-Sq	0.9887
Coeff Var	3.69055		

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variance Inflation
Intercept	1	-34.07559	0.12714	-268.01	<.0001	0
HoursStudied	1	2.85298	0.00787	362.35	<.0001	1.00048
PreviousScores	1	1.01843	0.00118	866.45	<.0001	1.00033
SleepHours	1	0.48056	0.01202	39.97	<.0001	1.00060
SampleQuestionPapersPracticed	1	0.19380	0.00711	27.26	<.0001	1.00056
Extra	1	0.61290	0.04078	15.03	<.0001	1.00080

Appendix 9: Stepwise Regression Output (final step)

Bounds on condition number: 1.0005, 16.005

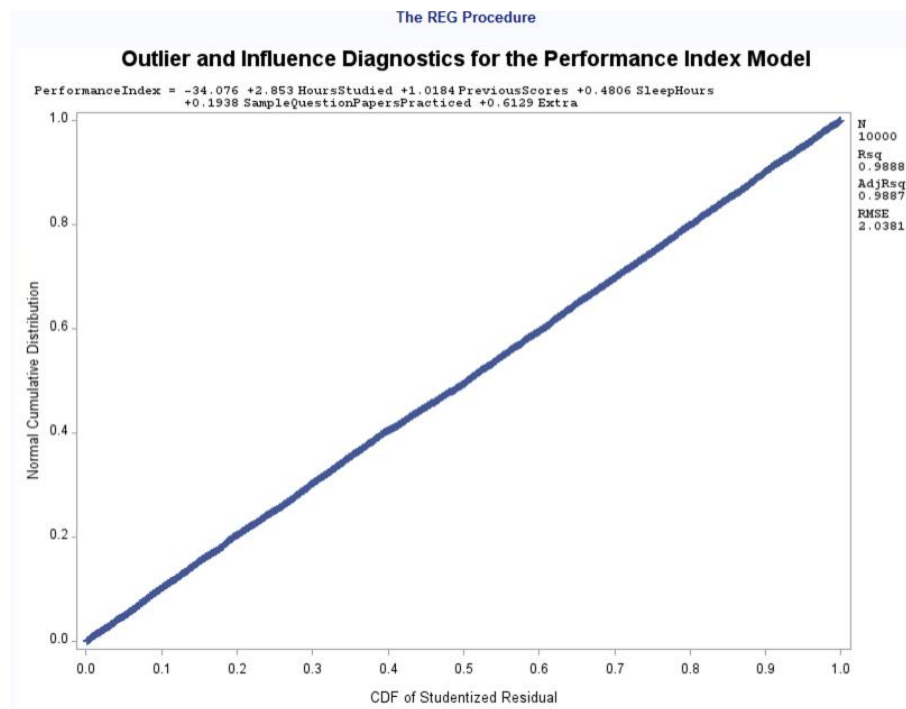
Stepwise Selection: Step 5

Variable Extra Entered: R-Square = 0.9888 and C(p) = 6.0000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	3649341	729868	175709	<.0001
Error	9994	41514	4.15384		
Corrected Total	9999	3690855			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	-34.07559	0.12714	298369	71829.6	<.0001
HoursStudied	2.85298	0.00787	545400	131300	<.0001
PreviousScores	1.01843	0.00118	3118438	750736	<.0001
SleepHours	0.48056	0.01202	6637.00854	1597.80	<.0001
SampleQuestionPapersPracticed	0.19380	0.00711	3085.97661	742.92	<.0001
Extra	0.61290	0.04078	938.25475	225.88	<.0001

Appendix 10: Diagnostic Plot (Normal Probability Plot of Studentized Residuals)



Appendix 11: Prediction Output for Sample Student

Compute Predictions for Performance Index

The REG Procedure
Model: MODEL1
Dependent Variable: PerformanceIndex

Output Statistics								
Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	95% CL Mean		95% CL Predict		Residual
1	.	74.5550	0.0357	74.4851	74.6250	70.5593	78.5507	.
2	65	63.1578	0.0488	63.0621	63.2535	59.1616	67.1540	1.8422
3	66	67.0913	0.0377	67.0174	67.1653	63.0956	71.0871	-1.0913
4	61	59.4091	0.0461	59.3187	59.4995	55.4130	63.4052	1.5909
5	61	62.8408	0.0391	62.7641	62.9175	58.8450	66.8367	-1.8408
6	69	69.8992	0.0584	69.7847	70.0138	65.9025	73.8959	-0.8992
7	84	84.3170	0.0550	84.2092	84.4248	80.3205	88.3136	-0.3170
8	73	72.4755	0.0438	72.3897	72.5613	68.4795	76.4715	0.5245
9	27	27.0624	0.0542	26.9561	27.1687	23.0659	31.0589	-0.0624
10	33	33.2186	0.0532	33.1142	33.3229	29.2221	37.2150	-0.2186